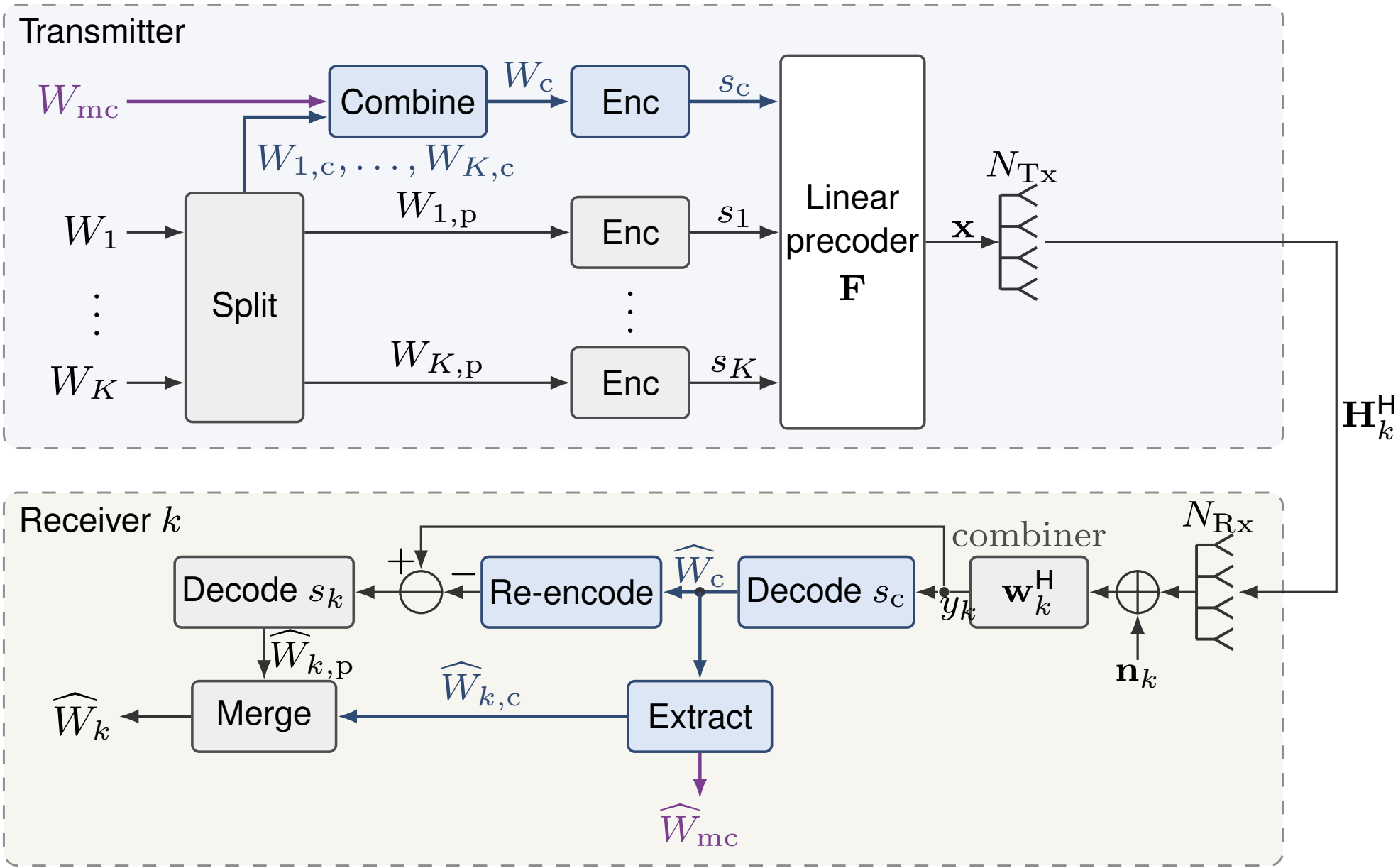

Transmitter
W_{mc}
Combine
W_{c}
Enc
s_{c}
$W_{1,\mathrm{c}}, \ldots, W_{K,\mathrm{c}}$
W_1
W_K
Split
$W_{1,\mathrm{p}}$
$W_{K,\mathrm{p}}$
Enc
Enc
s_1
s_K
Linear precoder $\mathbf{F}$
$\mathbf{x}$
N_{Tx}
$\mathbf{H}_k^{\mathsf{H}}$
Receiver k
N_{Rx}
combiner
$\mathbf{w}_k^{\mathsf{H}}$
y_k
$\mathbf{n}_k$
Decode s_k
Re-encode
Decode s_{c}
$\widehat{W}_{\mathrm{c}}$
$\widehat{W}_{k,\mathrm{p}}$
$\widehat{W}_{k,\mathrm{c}}$
Merge
Extract
$\widehat{W}_k$
$\widehat{W}_{\mathrm{mc}}$